

MAI THU NGUYEN

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Affiliation: University of Lausanne, Switzerland

I am an evolutionary biologist studying sexual systems, life-history strategies, and plant demography. My research combines fieldwork, experiments, and quantitative modelling to investigate how genetic, evolutionary, and ecological processes shape reproductive strategies and population dynamics.

Education

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| 2021–2026 | PhD in Ecology and Evolution — University of Lausanne, Switzerland
Supervisor: Prof. John Pannell
Thesis: The coexistence of males, females and hermaphrodites in the annual herb <i>Mercurialis annua</i> . |
| 2018–2021 | MSc in Systems Biology — ETH Zurich, Switzerland
Supervisor: Prof. Roland Regoes
Thesis: The relationship between fitness landscapes and the distribution of fitness effects of new mutations. |
| 2015–2018 | BSc in Bioscience — University of Heidelberg, Germany |

Other Research Experiences

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| 2021 | Research Assistant — Sandec, EAWAG/ETH Zurich
Project: Particle size as a driver of dewatering performance and its relationship to stabilization in fecal sludge. |
| 2017–2018 | Research Intern — BioQuant, University of Heidelberg
Project: Modelling metabolic and protein network responses to varying calcium input signals. |

Publications

- **Nguyen, M.T.** & Pannell, J.R. (2026). Geographic gradients in leaky sex expression and reproductive effort in a dioecious plant are consistent with selection during range expansion. *Journal of Evolutionary Biology*, voag044.
- **Nguyen, M.T.** & Pannell, J.R. (2025). The evolution and maintenance of trioecy with cytoplasmic male sterility. *Heredity*, 134: 1–9.
- **Nguyen, M.T.**, Martignier, T., & Pannell, J.R. (2024). Widespread male sterility and trioecy in *Mercurialis annua*: Its distribution, genetic basis, and estimates of morph-specific fitness components. *American Journal of Botany*, 111: e16429.
- Ward, B.J., **Nguyen, M.T.**, et al. (2023). Particle size as a driver of dewatering performance and its relationship to stabilization in fecal sludge. *Journal of Environmental Management*, 326: 116801.

Conferences & Presentations

- European Society for Evolutionary Biology Congress, Barcelona (2025) — Poster.
- Models in Population Dynamics, Ecology & Evolution Conference, Bilbao (2025) — Talk.
- International Botanical Congress, Madrid (2024) — Talk.
- Biology24, Zurich (2024) — Poster.
- European Meeting for PhD Students in Evolutionary Biology 28, Isle of Cumbrae, Scotland (2023) — Talk.
- Biology23, Geneva (2023) — Poster.
- European Society for Evolutionary Biology Congress, Prague (2022) — Poster.
- Biology22, Basel (2022) — Talk.

Teaching & Mentoring

Teaching	Botany — BSc annual lecture (2022-2025) Ecology and Evolution of Mediterranean Flora — MSc field course (2025)
Mentoring	Semester project — MSc student (2025) Experimental design — BSc group project (2022)

Skills

Computational: R, Python, Mathematica, SliM, Matlab
Fieldwork: Sex ratio estimation, population surveys, citizen science seed collection management
Statistical: Mixed models, comparative methods
Experimental: PCR, qPCR, immunohistochemistry, microscopy

Awards & Fellowships

Delegate of Vietnam, Asian Science Camp, Tsukuba, 2013 — selected as one of five national delegates representing Vietnam.

Outreach & Service

2025	Organizing Committee Member, Biology25 Conference — selected talks and organized Conference dinner.
2020–2021	Volunteer, AOZ Zurich — supported refugee integration.
2021	Program Developer, Silver Lining Vietnam — designed STEM outreach programs for rural schools

References

Prof. John Pannell — University of Lausanne — john.pannell@unil.ch
Prof. Roland Regoes — ETH Zurich — roland.regoes@env.ethz.ch